Q1

(a). The local market approach examines the effects of immigration by comparing labor market outcomes across geographic areas that receive different numbers of immigrants. This approach allows researchers to compare areas while controlling for nationwide economic trends.

(b) & (c).

Selection bias. Immigrants may choose to move in the areas with faster wage growth. Thus, researchers may overstate the positive wage effects or understate the negative wage effects.

Spillover effect. An inflow of immigrants into one city may initially lower wages, encouraging some native workers to relocate to other cities. This movement may equalize wages across cities and understate the negative wage effects.

Q2.

(a) $\sigma = 10$ (the coefficient is -10)

(b)

$$\ln\left(\frac{L_{SN}}{L_{SF}}\right) = \text{Constant} - 10 \times \ln\left(\frac{w_{SN}}{w_{SF}}\right)$$

$$\Rightarrow \ln\left(\frac{w_{SN}}{w_{SF}}\right) = \text{Constant} - \frac{1}{10} \times \ln\left(\frac{L_{SN}}{L_{SF}}\right)$$

$$\Rightarrow \Delta \ln\left(\frac{w_{SN}}{w_{SF}}\right) = -\frac{1}{10}\,\Delta \ln\left(\frac{L_{SN}}{L_{SF}}\right)$$

$\Delta \ln L_{SF} = 10\% \Rightarrow \Delta \ln\left(\frac{L_{SN}}{L_{SF}}\right) = -10\% \Rightarrow \Delta \ln\left(\frac{w_{SN}}{w_{SF}}\right) = 1\%$

Q3.

$$\frac{\Delta Y}{Y} = \frac{\Delta A}{A} + \frac{1}{3}\frac{\Delta K}{K} + \underbrace{\frac{1}{3}\frac{\Delta L_S}{L_S} + \frac{1}{3}\frac{\Delta L_U}{L_U}}$$

$$= \frac{1}{3}\left[(1-\lambda_F)\left(\frac{\Delta L_{SN}}{L_{SN}} + \frac{\Delta L_{UN}}{L_{UN}}\right) + \lambda_F \left(\frac{\Delta L_{SF}}{L_{SF}} + \frac{\Delta L_{UF}}{L_{UF}}\right)\right]$$

$\lambda_F = 0.13$; $\frac{\Delta L_{SF}}{L_{SF}} = 0.26$; $\frac{\Delta L_{UF}}{L_{UF}} = 0.26$

(a) $\frac{\Delta L_{SF}}{L_{SF}} = \frac{\Delta L_{UF}}{L_{UF}} = 0.13$

$\Rightarrow g_Y^{direct} = \lambda_F \left[\frac{1}{3} \frac{\Delta L_{SF}}{L_{SF}} + \frac{1}{3} \frac{\Delta L_{UF}}{L_{UF}} \right] \approx 1.13\%$

* $Y = \frac{1}{3} K^{-\frac{2}{3}} L_S^{\frac{1}{3}} L_U^{\frac{1}{3}} \Rightarrow g_Y = -\frac{2}{3} g_K + \frac{1}{3} g_{LS} + \frac{1}{3} g_{LU} = 0$ in the long run

When $g_Y = 0$, $g_K = g_Y$ (Steady State)

$0 = -\frac{2}{3} g_Y + \underbrace{\frac{1}{3} g_{LS} + \frac{1}{3} g_{LU}}_{\frac{1}{3}(0.13 \times (0.13 + 0.13))}$

λ_F: 0.13×0.26

$\Rightarrow g_Y = \frac{1}{2}(0.13 \times 0.26) = 0.169 = \boxed{16.9\%}$

(b) ① If $L_{SF} = L_{UF}$, $\frac{\Delta L_{SF}}{L_{SF}} + \frac{\Delta L_{UF}}{L_{UF}} = 13\%$ ($\frac{\Delta L_{SF}}{L_{SF}}$: 13%, $\frac{\Delta L_{UF}}{L_{UF}}$: 13%)

② If $L_{SF} < L_{UF}$, $\frac{\Delta L_{SF}}{L_{SF}} + \frac{\Delta L_{UF}}{L_{UF}} > 13\%$ because $\Delta L_{UF} > \Delta L_{SF}$ in (a) ($\frac{\Delta L_{SF}}{L_{SF}}$: 13%, $\frac{\Delta L_{UF}}{L_{UF}}$: 13%)

In case ①, the answer would be the same as (a) given $\frac{\Delta A}{A} = 0$

In case ②, $g_Y > 1.69\%$

Moreover, if skilled immigrants bring innovation, $\frac{\Delta A}{A} > 0 \Rightarrow g_Y > 1.69\%$